

Question:
Which of the following factors is associated with a decreased risk of Alzheimer's?

- Answer Choices:
- A. Being African or Hispanic American
 - B. A lower level of education
 - C. Living in a city
 - D. Being married
 - E. Having a family history of Alzheimer's
 - F. High intake of saturated fats
 - G. Older age
 - H. Eating fish
 - I. Being a smoker
 - J. Drinking alcohol regularly

Reasoning + Answer:

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High-level rationale:

- Consistently documented risk factors for Alzheimer’s disease include older age, lower educational attainment, Apo E 4 carrier status (not listed), diabetes, hypertension, obesity, depression, and a history of traumatic brain injury. Certain modifiable lifestyle factors (for example, regular exercise, balanced diet, social engagement) may be associated with lower risk, whereas smoking, high saturated fat intake, and heavy alcohol use are linked to increased vascular and neurodegenerative risk.

- Among the options:
- Older age (G) is strongly positively associated with Alzheimer’s risk.
- Lower education (B), especially early-life cognitive reserve, is associated with higher risk.
- A family history (E) increases risk, especially with early-onset forms.
- Smoking (I) is a well-established risk factor.
- High saturated fat intake (F) and heavy alcohol use (J) can contribute to vascular disease and neurotoxicity, increasing risk.
- Urban vs rural residence (C) is not a reliable, independent protective factor; some studies have found mixed or no significant effect.
- Marriage status (D) does not have a consistently protective effect; certain studies suggest possible deleterious effects of widowhood, while being married is not reliably protective overall.
- Being African or Hispanic American (A) is not inherently associated with decreased risk; if anything, these populations may have higher prevalence in some datasets, possibly due to measurement artifact, co-morbidities, or access-to-care differences, and they are not ‘decreased risk’ per se.
- High fish intake (H) is not an established protective factor; associations are mixed and do not reach the level of a reliable, causal decrease in risk.

Evaluating these, the best-supported protective factor among the choices is being married; however, the most broadly accurate and systematically supported ‘decreased risk’ factor is exercise and physical activity, which are not listed.

Correct answer: D. Being married

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<answer>

D. Being married

</answer>

